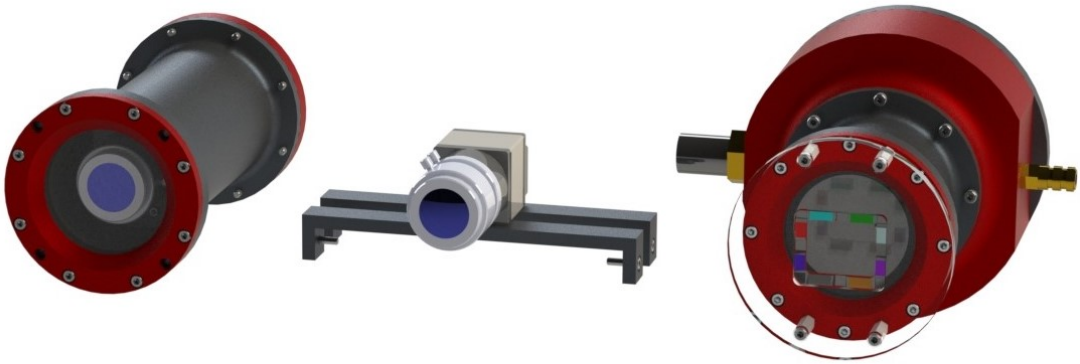
CU AUV

Cornell University Autonomous Underwater Vehicle Team

Spring 2021

Camera Enclosures



User Manual

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Contents

1 Abstract **2**

2 Aurora External Forward Facing Camera Enclosure **2**
2.1 Components . . . 2
2.2 Bill of Materials . . . 3
2.3 Interfacing . . . 3
2.4 Assembly Instructions . . . 3
2.5 Operating Instructions . . . 4

3 Aurora Internal Forward Camera-Lens Combination **4**
3.1 Components . . . 4
3.2 Bill of Materials . . . 5
3.3 Interfacing . . . 5
3.4 Assembly Instructions . . . 5
3.5 Operating Instructions . . . 5

4 Aurora External Downward Facing Camera Enclosure **6**
4.1 Components . . . 6
4.2 Bill of Materials . . . 6
4.3 Interfacing . . . 7
4.4 Assembly Instructions . . . 7
4.5 Operating Instructions . . . 8

5 Color Palette Fabrication **8**
5.1 Purpose . . . 8
5.2 Fabrication Instructions . . . 9
5.2.1 Option A - Colored Epoxy Resin . . . 9
5.2.2 Option B - Corrosion-Resistant Stickers . . . 10

1 Abstract

The camera enclosures must securely seal and enclose the camera-lens combination to prevent external fluid exposure. In particular, the camera-lens combinations, camera enclosures, and color palette must integrate methods of mounting to the body or frame. Respectively, the two forward facing and downward facing camera-lens combinations provide for a stereoscopic perspective and a distinct view cone direction.

2 Aurora External Forward Facing Camera Enclosure

2.1 Components

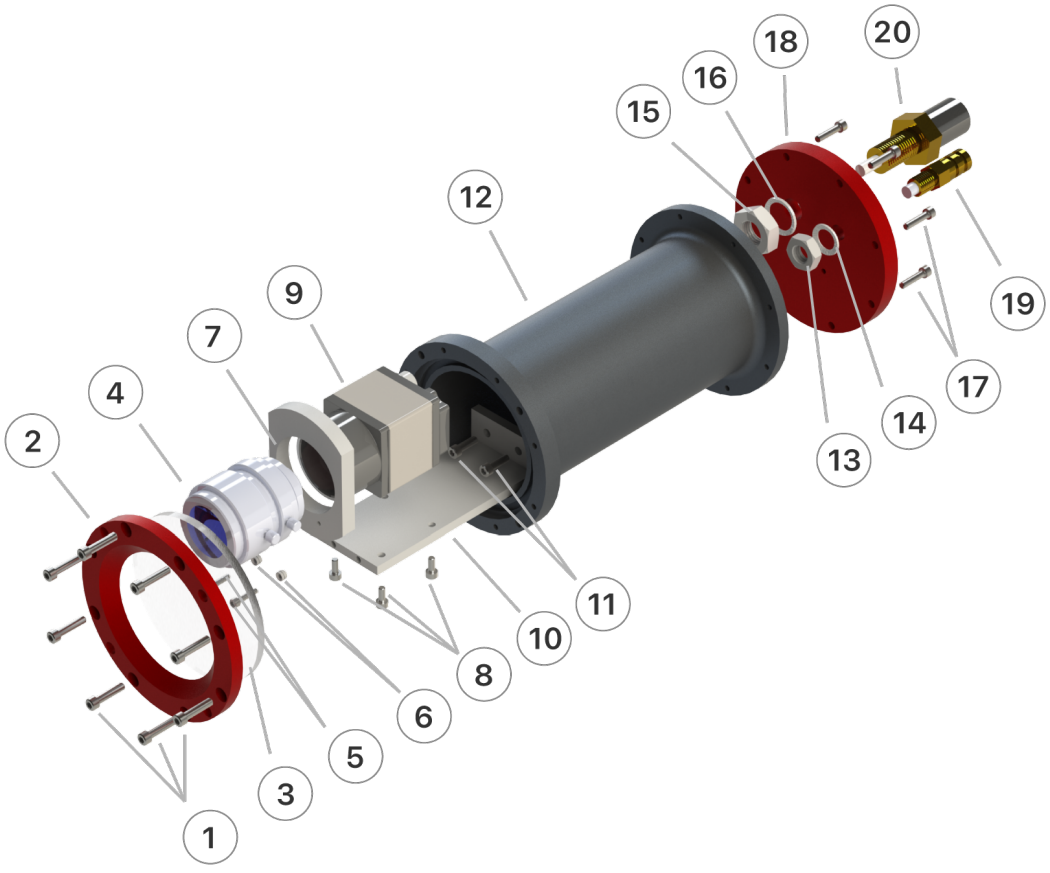


Figure 1: External Forward Facing Camera Enclosure Exploded View

2.2 Bill of Materials

Part #	Component	Qty.
1	#4-40 x 0.5” Socket Head Cap Screw	8
2	Camera Enclosure Flange	1
3	2-1/2” x 1/8” Borosilicate Glass Disk	1
4	NMV-4WA Lens	2
5	#2-56 x 0.3125” Socket Head Cap Screw	2
6	#2-56 x 0.1” Brass Heat-Set Inserts	2
7	Camera Enclosure Bulkhead	1
8	M2.5 x 0.45 x 6 mm Socket Head Cap Screw	4
9	UI-5140CP Camera	1
10	External Forward Facing Camera Enclosure Bracket	1
11	#4-40 x 0.375” Socket Head Cap Screw	2
12	External Forward Facing Camera Enclosure Hull	1
13	5/16”-24 Nut	1
14	5/16” Belleville Washer	1
15	7/16”-20 Nut	1
16	7/16” Belleville Washer	1
17	#4-40 x 0.5” Socket Head Cap Screw	8
18	External Forward Facing Camera Enclosure Endcap	1
19	HUMG-2	1
20	MCBH-8	1
-	Size 139 O-Ring	2

2.3 Interfacing

The Aurora External Forward Facing Camera Enclosure will interface with the Aurora frame, mounting using 4 brackets attaching to both the #4 clearance holes on the flange and #4-40 tapped holes on the hull.

2.4 Assembly Instructions

1. Use a set insert iron to insert (5) the brass heat set insert into (10) the plastic cantilevered bracket.

2. Mount (9) the UI-5140CP Camera onto (10) the plastic cantilevered bracket using (8) M2.5 x 0.45 x 6 mm Socket Head Cap Screws.

3. Place (7) the bulkhead around the cylindrical portion of (9) the UI-5140 CP Camera. Align the clearance hole on (7) the bulkhead with (5) the brass heat set insert [Instruction 1]. Then use (5) #2-56 x 0.3125” Socket Head Cap Screws to mount (7) the bulkhead in place.

4. Attach (4) the Navitar NMV-4WA Lens to (9) the UI-5140CP Camera.
5. Attach the camera-bracket combination [Instruction 1-4] to (18) the endcap using (11) #4-40 x 0.375" Socket Head Cap Screws.
6. Attach (19) the HUMG-2 to (18) the endcap using (13) a 5/16"-24 Nut and (14) a 5/16" Belleville Washer. Attach (20) the MCBH-8 to (18) the endcap using (15) a 7/16"-20 Nut and (16) a 7/16" Belleville Washer. Then wire the cameras with the SEACON connectors accordingly.
7. Place the current assembled enclosure [Instructions 1-6] into (12) the hull. Place a Size 139 O-Ring into the groove on the back side of the hull and connect the two parts using (17) #4-40 x 0.5" Socket Head Cap Screws.
8. Place another Size 139 O-Ring on the front side of the hull. Then place (3) 2-1/2" x 1/8" Borosilicate Glass Disk in the step down slot on the hull. Align (2) the flange and connect the flange to the assembly using (1) #4-40 x 0.5" Socket Head Cap Screws.

2.5 Operating Instructions

- Ensure the enclosure is properly sealed before use.
- Use sealing grease on the O-Rings.
- Avoid touching the lens of the camera.

3 Aurora Internal Forward Camera-Lens Combination

3.1 Components

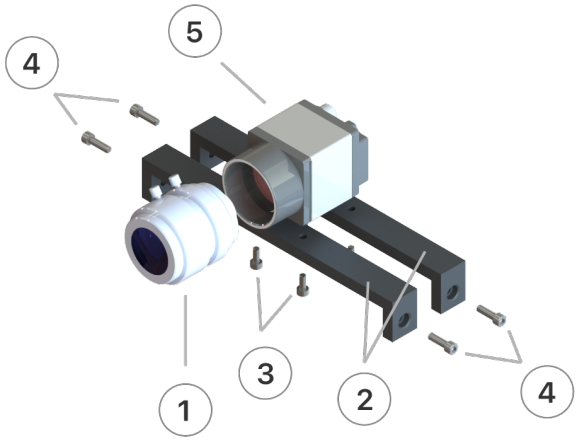


Figure 1: Internal Forward Camera-Lens Combination Exploded View

3.2 Bill of Materials

Part #	Component	Qty.
1	NMV-4WA Lens	2
2	Internal Forward Camera-Lens Bracket	2
3	M2.5 x 0.45 x 6 mm Socket Head Cap Screw	4
4	#4-40 x 0.375" Socket Head Cap Screw	4
5	UI-5140CP Camera	1

3.3 Interfacing

The Aurora Internal Forward Camera-Lens Combination will interface with the Aurora racks which is located in the UHPV (Upper Hull Pressure Vessel), mounting using four #4-40 x 0.375" Socket Head Cap Screws. The two brackets of this assembly will attach to the truss of the racks and will be located above the Jetson.

3.4 Assembly Instructions

1. Mount one of (2) the brackets to (5) the UI-5140CP Camera using two (3) M2.5 x 0.45 x 6 mm Socket Head Cap Screw.
2. Repeat [Instruction 1] for the other (2) bracket.
3. Attach (1) the Navitar NMV-4WA Lens to the (3) UI-5140CP Camera.
4. Place the current assembly [Instructions 1-2] on the truss of the racks.
5. Align the counterbore holes of (2) the brackets to the mounting holes located on the truss.
6. Mount using (4) #4-40 x 0.375" Socket Head Cap Screws. Then wire the cameras accordingly.

3.5 Operating Instructions

- Ensure the UHPV is properly sealed before use.
- Avoid touching the lens of the camera.
- Mount one bracket at a time to the camera.
- Ensure assembly does not interfere with the electronic boards or the Jetson.
- Ensure the viewcone of the camera is not obstructed by the flange of the UHPV. Adjust the cameras and remount if needed to ensure viewcone is co-linear with that of the camera in the External Forward Facing Camera Enclosure.

4 Aurora External Downward Facing Camera Enclosure

4.1 Components

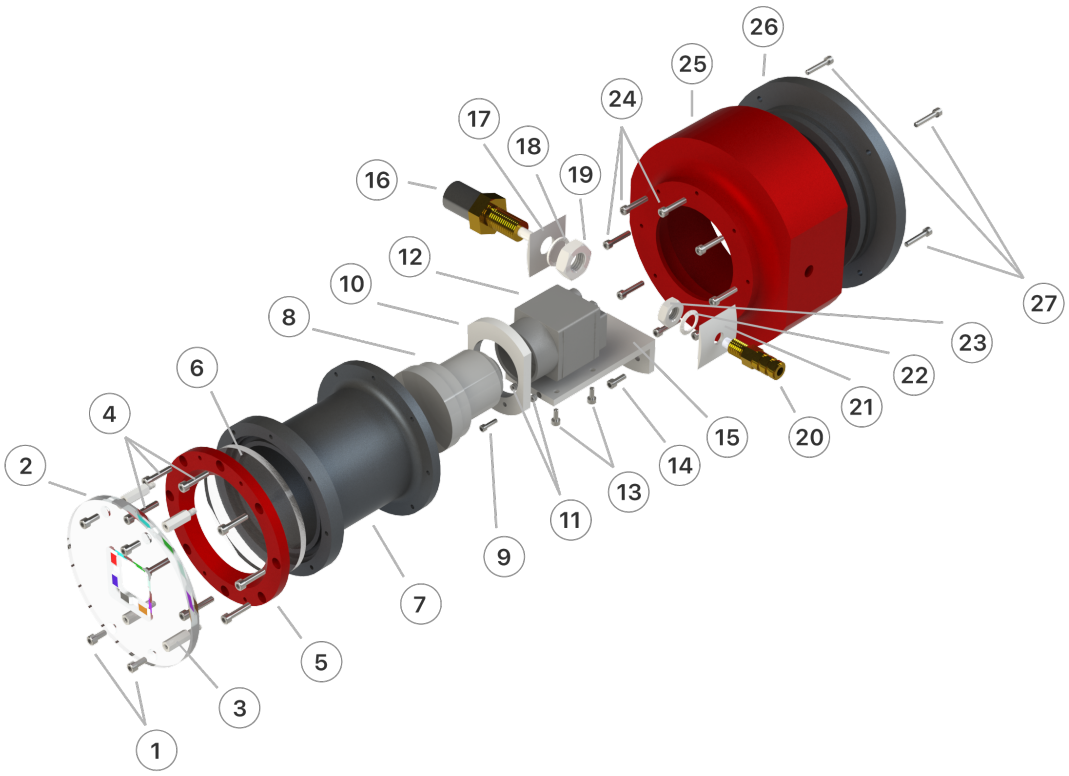


Figure 1: External Downward Facing Camera Enclosure Exploded View

4.2 Bill of Materials

Part #	Component	Qty.
1	#4-40 x 0.25” Socket Head Cap Screw	4
2	Color Palette [Refer to Section 5]	1
3	#4-40 Male-Female Threaded Hex Standoff	4
4	#4-40 x 0.5” Socket Head Cap Screw	8
5	Camera Enclosure Flange	1
6	2-1/2” x 1/8” Borosilicate Glass Disk	1

7	External Downward Facing Camera Enclosure Hull	1
8	Kowa LM3NCM Lens	1
9	#2-56 x 0.3125” Socket Head Cap Screw	2
10	Camera Enclosure Bulkhead	1
11	#2-56 x 0.1” Brass Heat-Set Inserts	2
12	UI-5250CP Camera	1
13	M2.5 x 0.45 x 6 mm Socket Head Cap Screw	4
14	#4-40 x 0.3125” Socket Head Cap Screw	2
15	External Downward Facing Camera Enclosure Bracket	1
16	MCBH-8	1
17	7/16”-20 Fixer	1
18	7/16” Belleville Washer	1
19	7/16”-20 Nut	1
20	HUMG-2	1
21	5/16”-24 Fixer	1
22	5/16” Belleville Washer	1
23	5/16”-24 Nut	1
24	#4-40 x 0.5625” Socket Head Cap Screw	8
25	External Downward Facing Camera Enclosure Endcap Hull	1
26	External Downward Facing Camera Enclosure Endcap	1
27	#4-40 x 0.625” Socket Head Cap Screw	4
-	Size 139 O-Ring	2
-	Size 152 O-Ring	1

4.3 Interfacing

The Aurora External Downward Facing Camera Enclosure will interface with the Aurora frame, mounting using 2 brackets attaching to four #4-40 tapped holes on the hull.

4.4 Assembly Instructions

1. Use a set insert iron to insert (11) the brass heat set insert into (15) the plastic cantilevered bracket.
2. Mount (12) the UI-5250CP Camera onto (15) the plastic cantilevered bracket using (13) M2.5 x 0.45 x 6 mm Socket Head Cap Screws.
3. Place (10) the bulkhead around the cylindrical portion of (12) the UI-5250 CP Camera. Align the clearance hole on (10) the bulkhead with (11) the brass heat set insert [Instruction 1]. Then use (9) #2-56 x 0.3125” Socket Head Cap Screws to mount (10) the bulkhead in place.

4. Attach (8) the Kowa LM3NCM Lens to (12) the UI-5250CP Camera.
5. Attach the camera-bracket combination [Instruction 1-4] to (26) the endcap using (14) #4-40 x 0.3125" Socket Head Cap Screws.
6. Attach (16) the MCBH-8 to (25) the endcap hull using a cylindrical (17) 7/16"-20 Fixer, (18) a 7/16" Belleville Washer, and (19) a 7/16"-20 Nut. Attach (20) the HUMG-2 to (25) the endcap hull using a cylindrical (21) 5/16"-24 Fixer, (22) a 5/16" Belleville Washer, and (23) a 5/16"-24 Nut. Then wire the cameras with the SEACON connectors accordingly.
7. Place a Size 152 O-Ring on the bore seal slot on the (26) endcap. Then mount the camera-endcap [Instructions 1-5] to the endcap hull [Instruction 6] using (27) #4-40 x 0.625" Socket Head Cap Screws.
8. Place the current assembled enclosure [Instructions 1-7] into (7) the hull. Place a Size 139 O-Ring into the groove on the back side of the hull and connect the two parts using (24) #4-40 x 0.5625" Socket Head Cap Screws.
9. Place another Size 139 O-Ring on the front side of the hull. Then place (6) 2-1/2" x 1/8" Borosilicate Glass Disk in the step down slot on the hull. Align (5) the flange and connect the flange to the assembly using (4) #4-40 x 0.5" Socket Head Cap Screws.
10. Align and mount (3) the Male-sided #4-40 Threaded Hex Standoffs to (5) the flange.
11. Mount (2) the Color Palette to the (3) the Female-sided #4-40 Threaded Hex Standoffs using (1) #4-40 x 0.25" Socket Head Cap Screws.

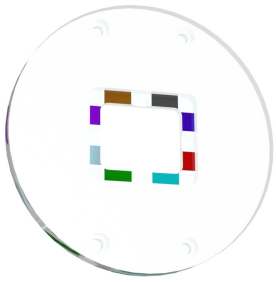
4.5 Operating Instructions

- Ensure the enclosure is properly sealed before use.
- Use sealing grease on the O-Rings.
- Avoid touching the lens of the camera.

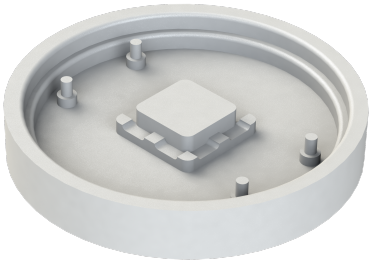
5 Color Palette Fabrication

5.1 Purpose

The color perception palette is a newly implemented feature and utilizes a 3D-printed ABS plastic mold to allow clear epoxy resin to form the shape desired for the palette. Most notably, the center rectangle forms an outer ring of eight colors in the downward camera's viewcone. In particular, the colors of known RGB values are manually chosen to best fit the ability to translate the underwater colors to surfaced colors.



(a) Color Palette



(b) Mold

Figure 1: Aurora External Downward Enclosure Color Perception Palette

5.2 Fabrication Instructions

5.2.1 Option A - Colored Epoxy Resin

1. Prepare Clear Epoxy Resin and use the mold (Figure 1 (b)) to make the structure of the Color Palette (Figure 1 (a)).
2. Spray the mold with mold release. Then pour the resin [Instruction 1] into the mold.
3. Let the mold cure. Once done, remove the Color Palette from the mold.
4. Determine the RGB colors to be used for the color palette. Prepare eight colors of epoxy resin using a calculated # of drops of coloring or amount of color pigment powder.
5. Create structure for the center opening and cover this with wax paper, painters tape, or any material that doesn't stick well with resin. Place this structure inside the center opening then cure the resin [Instruction 4] inside each of the eight slots on the color palette.
6. Let the mold cure. Once done, remove the center opening structure created in [Instruction 5].

Operating Instructions

- Give the epoxy time to cure.
- Use a cover to reduce epoxy resin dust exposure.
- Use dimensions of each slot to calculate slot volume.
- Calculate using volume the amount of epoxy and coloring pigment to use.

5.2.2 Option B - Corrosion-Resistant Stickers

1. Prepare Clear Epoxy Resin and use the mold (Figure 1 (b)) to make the structure of the Color Palette (Figure 1 (a)).
2. Spray the mold with mold release. Then pour the resin [Instruction 1] into the mold.
3. Let the mold cure. Once done, remove the Color Palette from the mold.
4. Determine the RGB colors to be used for the color palette. Prepare eight colors of non-corrosive stickers. Then place the stickers in their respective slot locations.

Operating Instructions

- Give the epoxy time to cure.
- Use a cover to reduce epoxy resin dust exposure.
- Use dimensions of each slot to prepare the right sticker size.
- Add a layer of clear epoxy resin on each slot if stickers continually fall off or get worn easily after use.